

CS 541P: IoT systems and the Cloud

Course Code	CS 541P
Course Name	IoT systems and the Cloud
L-T-P-C	1-0-3-3
Prerequisites	IC161- Applied Electronics or equivalent, IC250 - Programming and Data Structure Practicum or equivalent with Consent of Instructor
Intended for	B.Tech. (EE. & CSE) /MS/MTech/PhD Distribution Semester: Elective for B.Tech. TII/IV year, MS, M.Tech., Ph.D.
Approval	19th BoA (Nov 2017); 15th Senate (Feb 2018)

Course Contents

- **Module 1: Hardware components of IoT systems:** Introduction to microcontrollers including PIC, Arduino, Raspberry Pi; interfacing sensors and actuators with microcontrollers, building breadboard circuits, using IoT CAD tools; performance characteristics, drift, measurement techniques, packaging. [4 Lectures + 12 lab hours]
- **Module 2: Networking of devices:** Device-to-device communication; networking protocols and architectures; scheduling and routing; wired and wireless (Bluetooth, Zigbee, NFC) sensor networks; communication technologies like LoRA, SigFox, Cellular IoT; edge analytics and sensor control in networks. [3 Lectures + 9 lab hours]
- **Module 3: Getting Data Ready for Analytics in the Cloud:** Introduction to cloud service models, pre-processing input streams of IoT data, storage in the cloud - message queues, distributed file systems and distributed databases. [3 Lectures + 9 lab hours]
- **Module 4: Data Analytics in the Cloud:** Computing paradigms - map-reduce and its extensions to resilient distributed datasets, concepts in complex event processing (CEP), algorithms for mining data streams, Lambda Architecture. [4 Lectures + 12 lab hours]

Text Books

1. Donald Norris, **The Internet of Things**, McGraw Hill Education, 2015.
2. Nathan Marz and James Warren, **Big Data: Principles and best practices of scalable realtime data systems**, Manning Publications, 2015.

References

1. Upton and Halfacree, **Raspberry Pi User Guide**, Wiley, 2014.
2. Robert Faludi, **Building Wireless Sensor Networks**, O'Reilly Media, 2011.
3. Jure Leskovec, Anand Rajaraman and Jeffrey D. Ullman, **Mining of Massive Datasets**, v2.1 online edition, 2015.
4. Research papers on Resilient Distributed Datasets, Kafka, HBase, etc.